Monthly crime trends: a global STL-style decomposition

Andrew P. Wheeler

What this project estimates

This is a descriptive decomposition of monthly crime counts for every city in the RTCI national sample. Rates are annualized from monthly counts: a monthly count Y in population N is displayed as 12 times Y divided by N, multiplied by 100,000. This puts homicide on the familiar annual rate scale rather than below one.

The project uses one aggregated-binomial mixed model. For crime type (c), city (i), and month (t),

$$Y_{ict} \sim \mathrm{Binomial}(N_i, p_{ict}),$$

$$\mathrm{logit}(p_{ict}) = \alpha_c + f_c(t) + s_c(m_t) + b_i, \qquad b_i \sim N(0, \sigma_b^2).$$

Here alpha is the crime-specific intercept, f is a smooth crime-specific time trend, s is a cyclic month-of-year effect, and b is the city random intercept. The global prediction excludes b; the city prediction includes it.

For the requested city-by-crime-by-month overdispersion term, the observed probability is stabilized as

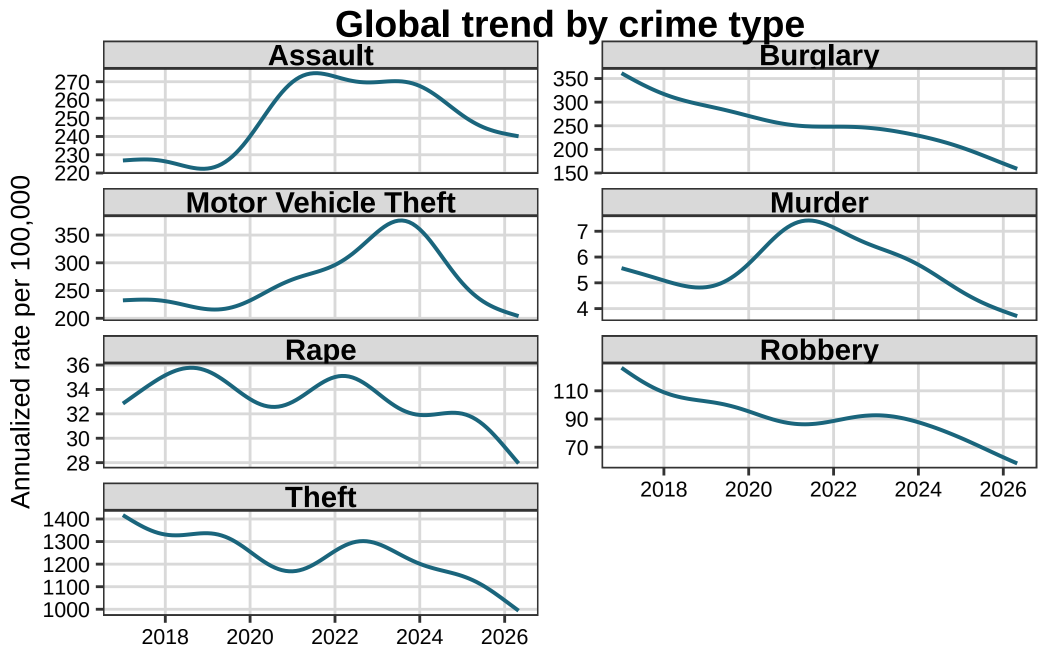
$$\tilde{p}_{ict} = \frac{Y_{ict} + 0.5}{N_i + 1}, \qquad o_{ict} = \mathrm{logit}(\tilde{p}_{ict}) - \mathrm{logit}(p_{ict}) - \bar{o}_{ic},$$

where the barred o term is the population-weighted mean within city i and crime type c. Therefore the overdispersion term is centered at zero for every city and crime type. It is not a Pearson residual. The global residual is constructed analogously from the global prediction and centered within crime type.

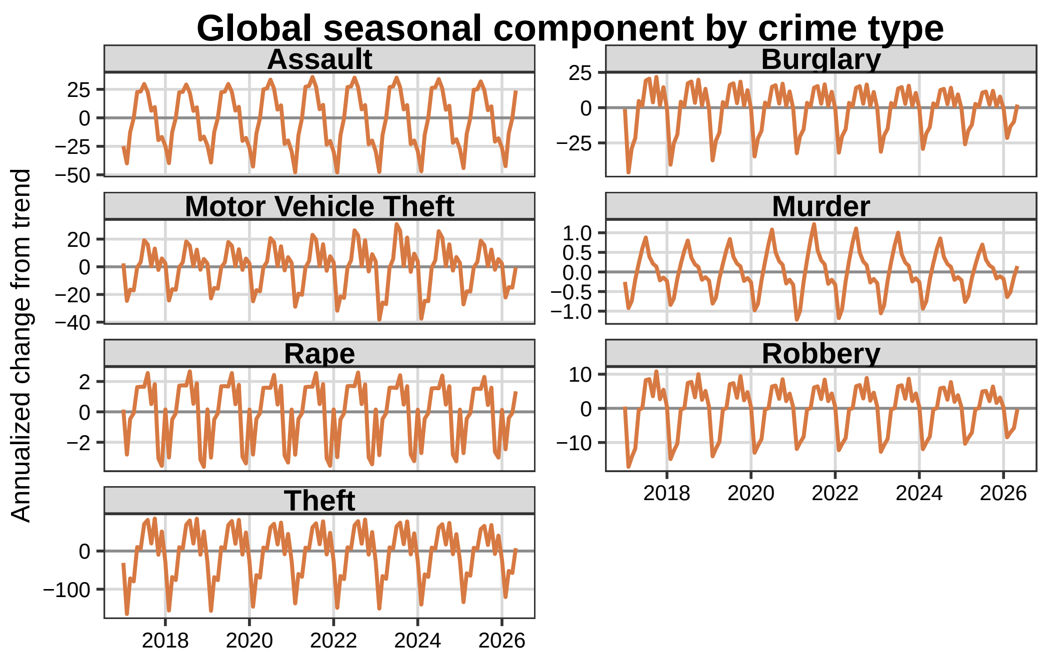
Data coverage

The deliverable contains 458,392 valid city-month-crime observations from 586 cities. There is no population threshold.

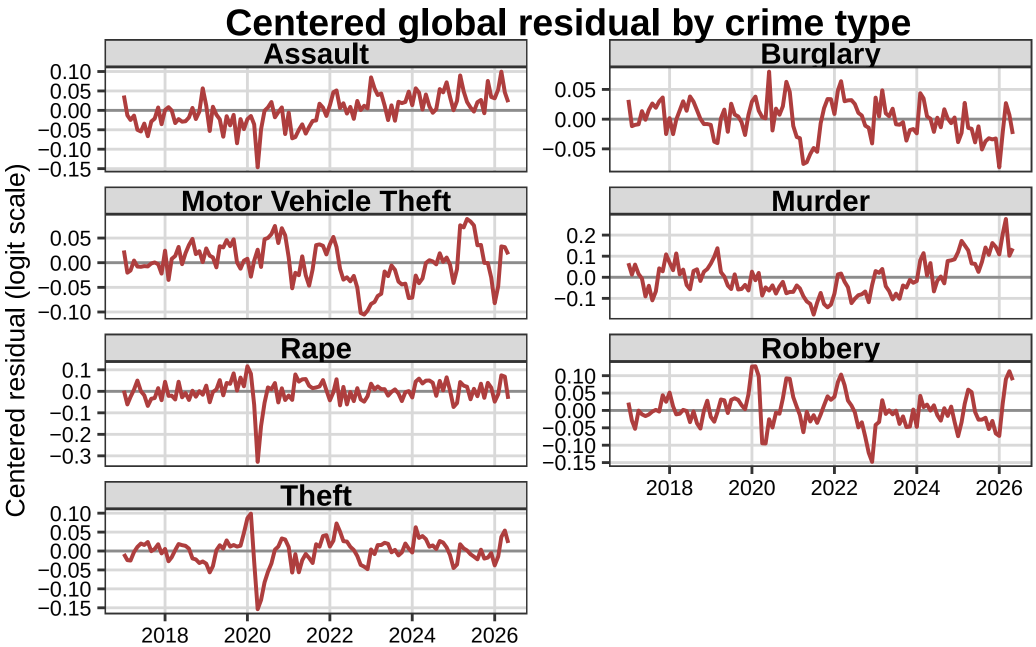
Global trend



Global seasonal component



Global residual



Reading the trend and residual together

The trend is a smooth long-run level, while the residual is a short-run departure from that level. Thus a declining murder trend can coexist with positive recent residuals: recent observations can be above the currently declining baseline even while the baseline itself remains below earlier years.